

Windows Server Storage Spaces Configurations Cheat Sheet

Simple (No Resiliency)

What it is: Data is written across all available drives without any redundancy or protection.

How it works: Data is striped across multiple drives to improve performance, but if any drive fails, all data is lost.

Used for:

- Temporary storage or scratch space
- Non-critical data where performance is more important than protection
- Testing environments
- When you have separate backup solutions

Minimum drives: 1 drive **Capacity:** 100% of total drive space

Mirror (Two-way or Three-way)

What it is: Data is duplicated across multiple drives for redundancy and fault tolerance.

How it works:

- **Two-way mirror:** Each piece of data is written to two drives
- **Three-way mirror:** Each piece of data is written to three drives

Used for:

- Critical business data
- Database files
- System drives
- High-availability applications
- When data protection is the top priority

Minimum drives:

- Two-way: 2 drives
- Three-way: 3 drives

Capacity:

- Two-way: 50% of total drive space
- Three-way: 33% of total drive space

Fault tolerance:

- Two-way: Can lose 1 drive
 - Three-way: Can lose 2 drives
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Parity (Single or Dual)

What it is: Uses mathematical calculations to provide fault tolerance with better space efficiency than mirroring.

How it works:

- **Single parity:** Calculates and stores parity information that can rebuild data if one drive fails
- **Dual parity:** Uses two sets of parity calculations for protection against two drive failures

Used for:

- Archival storage
- Large file storage (videos, backups)
- Read-heavy workloads
- When you need good capacity with some protection

Minimum drives:

- Single parity: 3 drives
- Dual parity: 7 drives

Capacity:

- Single parity: ~67-90% depending on number of drives
- Dual parity: ~60-80% depending on number of drives

Fault tolerance:

- Single parity: Can lose 1 drive
- Dual parity: Can lose 2 drives

Note: Slower write performance than Simple or Mirror configurations

Quick Comparison

Configuration	Min Drives	Fault Tolerance	Capacity Efficiency	Performance	Best For
Simple	1	None	100%	High read/write	Temporary data
Two-way Mirror	2	1 drive failure	50%	High read/write	Critical data
Three-way Mirror	3	2 drive failures	33%	High read/write	Mission-critical
Single Parity	3	1 drive failure	67-90%	Good read, slow write	Archives
Dual Parity	7	2 drive failures	60-80%	Good read, slow write	Large archives

Tips for Choosing

- **Performance priority:** Use Simple (with backups) or Mirror
- **Maximum protection:** Use Three-way Mirror
- **Balance of protection and capacity:** Use Single Parity
- **Large storage with protection:** Use Dual Parity
- **Always consider:** Your backup strategy, workload type, and budget